SEMESTER 2 EXAMINATION 2024/25

ADVANCED MACHINE LEARNING

Duration 120 mins (2 hours)

This paper is a WRITE-ON examination paper.

You ***must*** write your Student ID on this Page and must not write your name anywhere on the paper.

All answers should be written within the designated boxes in this examination paper and sufficient space is provided for each question.

If, for some reason, space is required to complete or correct an answer to a question, use the “Additional Space” provided on the facing or adjacent page to the question. Clearly indicate which question the answer corresponds to.

No credit will be given for answers presented elsewhere and without clear indication of to what question they correspond. Blue answer books may be used for scratch; they will be discarded without being looked at.

Answer all parts of the question in section A (40 marks)
and ALL three questions from section B (20 marks each)

Student ID:

Question	Mark	*Arithmetic checked*	*Double Marked*
A1	/40		
B2	/20		
B3	/20		
B4	/20		
Total:	/100		

University approved calculators MAY be used.

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

14 page examination paper

 Version 1

Section A

A 1

(a) In classification, what is the effect of increasing the number of training examples on generalisation performance? Explain why this is. [5 marks]

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(b) Briefly describe the random forest algorithm **and** explain why it is effective. [5 marks]

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(c) Show that an empirical covariance matrix, $\mathbf{C}$, can be written as $\mathbf{X}\mathbf{X}^{\mathsf{T}}$ and hence prove that its eigenvalues are non-negative. [5 marks]

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(d) Prove that the set of positive semi-definite matrices forms a convex set. [5 marks]

$\overline{5}$

(e) In gradient descent (in one dimension) we would update the weight by

$$w^{(t+1)} = w^{(t)} - r\,L'(w^{(t)})$$

assume that $L(w) = w^2$, write down a closed form expression for $w^{(t)}$ in terms of the initial state $w^{(0)}$ and explains what happens when $r > 1$. [5 marks]

$\overline{5}$

(f) For the linear programming problem of maximising $\boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}$ subject to the constraints $\mathbf{M}\,\boldsymbol{x} = \boldsymbol{b}$ write down a Lagrangian $L(\boldsymbol{x}, \boldsymbol{\alpha})$ where $\boldsymbol{\alpha}$ is a set of Lagrange multipliers. By rewriting the Lagrangian in its dual form provide an equivalent minimisation problem for the Lagrange multipliers.

[5 marks]

$\overline{5}$

(g) Show that the beta distribution $\mathrm{Beta}(p|a,b) = p^{a-1}\,(1-p)^{b-1}/B(a,b)$ is a conjugate prior to the binomial likelihood $\mathrm{Binom}(k|n,p) = \binom{n}{k} p^k\,(1-p)^{n-k}$. Derive update equations for the parameters of the posterior distribution after observing k successes and $n-k$ failures. [5 marks]

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(h) Show that if X and Y are independent random variables then the entropy, $H_{(X,Y)}$ for the two random variables is equal to the sum of the entropies for X and Y. [5 marks]

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End of question A1

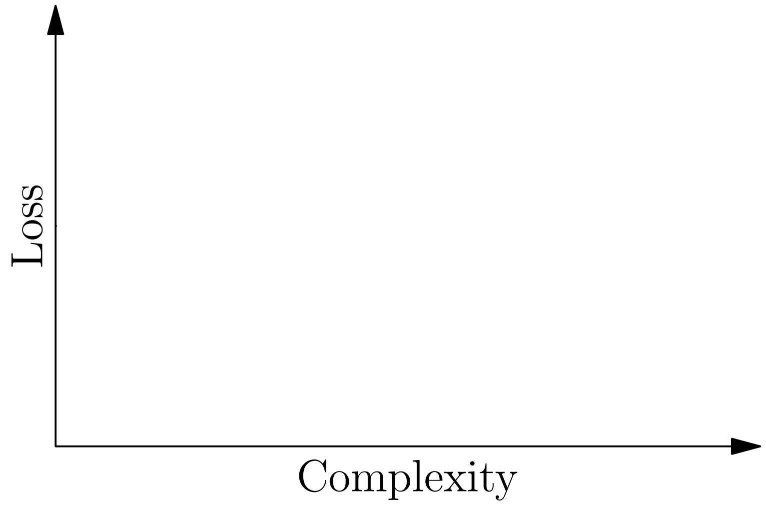
(a) $\frac{}{5}$ (b) $\frac{}{5}$ (c) $\frac{}{5}$ (d) $\frac{}{5}$ (e) $\frac{}{5}$ (f) $\frac{}{5}$ (g) $\frac{}{5}$ (h) $\frac{}{5}$ Total $\frac{}{40}$

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Section B

B 2

(a) On the axes given sketch the typical behaviour of the training and generalisation errors as a function of the models complexity (e.g the number of parameter). [5 marks]



$\overline{5}$

(b) Explain the training and generalisation loss in terms of the bias-variance dilemma. [5 marks]

$\overline{5}$

(c) Explain how in the Bayesian framework the model avoids overfitting. [5 marks]

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(d) Explain how hyper-parameter tuning can result in over-fitting, and how we can avoid this in Bayesian inference [5 marks]

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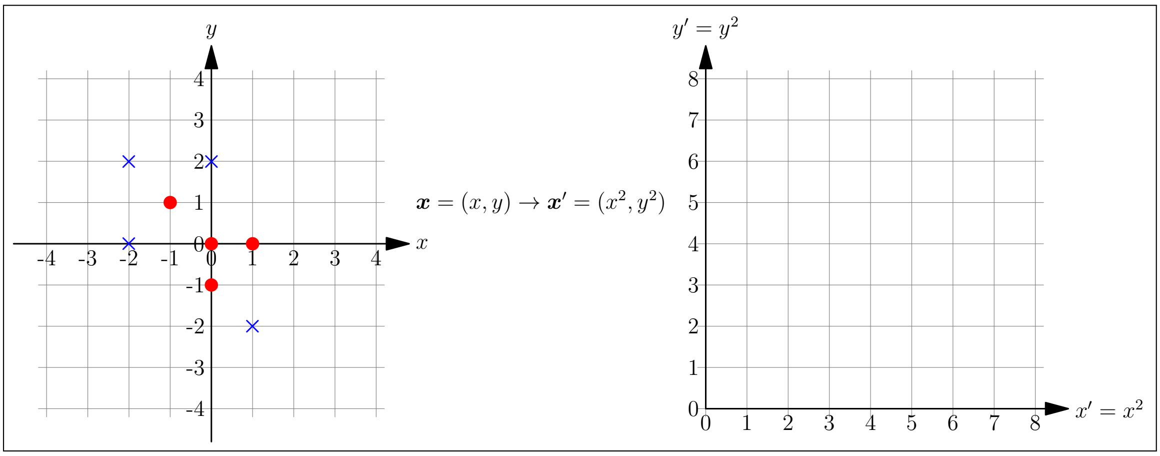
End of question B2

(a) $\frac{}{5}$ (b) $\frac{}{5}$ (c) $\frac{}{5}$ (d) $\frac{}{5}$ Total $\frac{}{20}$

Additional space. Do not use unless necessary. Clearly mark corresponding question.

B 3

(a) Show how the data points $\{\boldsymbol{x}_i = (x_i, y_i) | i = 1, 2, \ldots\}$ shown below transform under the mapping $\boldsymbol{x} = (x_i, y_i) \to \boldsymbol{x}' = (x_i^2, y_i^2)$ and sketch the position of the maximal margin dividing plane in the new feature space. [5 marks]



(b) Explain the kernel trick. [5 marks]

(c) Define what it means for a kernel to be positive semi-definite (giving different properties a positive semi-definite kernel satisfies). Explain why it is necessary for the kernel function of an SVM to be positive semi-definite. [5 marks]

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(d) Using the fact that any positive semi-definite kernel can be decomposed as

$$K(\boldsymbol{x}, \boldsymbol{y}) = \sum_i \phi_i(\boldsymbol{x})\, \phi_i(\boldsymbol{y})$$

show that the product of two kernel functions is positive semi-definite. [5 marks]

$\overline{}$ 5

End of question B3

(a) $\frac{}{5}$ (b) $\frac{}{5}$ (c) $\frac{}{5}$ (d) $\frac{}{5}$ Total $\frac{}{20}$

Additional space. Do not use unless necessary. Clearly mark corresponding question.

B 4

(a) Explain why Monte Carlo techniques are often used to solve Bayesian inference problems. [10 marks]

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(b) Briefly describe in words the use of the MCMC algorithm in Bayesian inference. [10 marks]

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10

End of question B4

(a) $\frac{}{10}$ (b) $\frac{}{10}$ Total $\frac{}{20}$

Additional space. Do not use unless necessary. Clearly mark corresponding question.